PATHOLOGY

Renal Stones & Cystic Diseases of the Kidney — II

Pathology GUE Block · MEDucation by Mattin Majid

Continued from Part I: this lecture's two tubular themes were primary tubular disease and tubulointerstitial disease. Renal stones and the cystic kidney diseases below are grouped separately in the lecture, under their own headings.

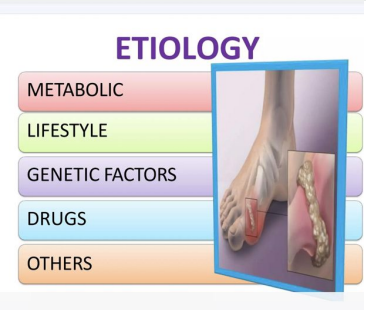
BEFORE YOU READ — ANSWER FROM MEMORY

1. Which stone type is radiolucent, forms in alkaline urine, and is linked to urease-positive organisms like Proteus?
2. What single microscopic feature separates multicystic renal dysplasia from the other cystic kidney diseases?
3. Between PKD1 and PKD2 mutations, which is more common and which causes earlier renal failure?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Renal stones

Terminology	**Nephrolithiasis** = renal stone disease; **urolithiasis** = presence of stones anywhere in the urinary system. Stones (calculi) form from the kidney to the bladder by crystallisation of substances excreted in urine.
Epidemiology	Incidence is **increased in men**.
Etiology categories	Metabolic, lifestyle, genetic factors, drugs, others *(lecture gives these as category headings only, without further detail under each).*



Etiology categories (lecture slide)

The four stone types — discriminator

Stone type	Frequency	Composition	Radiographic	Cause / association
Calcium	80–85% of urinary stones	Calcium oxalate or calcium phosphate, or both	**Radiopaque**	Increased intestinal Ca absorption; increased primary renal Ca excretion; hypercalcemia (hyperparathyroidism, malignancy — osteolytic mets or ectopic PTH —, sarcoidosis, vitamin D intoxication, milk-alkali syndrome)
Ammonium magnesium phosphate (struvite)	2nd most common	—	**Radiolucent**	Formed in alkaline urine caused by ammonia-producing (**urease-positive**) organisms, e.g. *Proteus vulgaris*. Can form large **staghorn** stones.
Uric acid	—	—	*not stated*	Hyperuricemia in ~half of patients — secondary to gout, or to increased cellular turnover (leukemias, myeloproliferative disorders)
Cystine	—	—	*not stated*	Almost always associated with **cystinuria** (genetically determined aminoaciduria)

The lecture gives frequency and radiographic appearance only for calcium and struvite stones — left blank above rather than filled in from outside knowledge.

2 Cystic diseases of the kidney

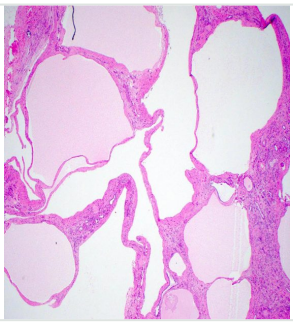
#	Group	Entities
1	Polycystic kidney disease	a. Autosomal-dominant (adult) PKD b. Autosomal-recessive (infantile) PKD
2	Multicystic renal dysplasia	—
3	Medullary cystic diseases	a. Medullary sponge kidney b. Uremic medullary cystic disease (nephronophthisis)
4	Acquired (dialysis-associated) cystic disease	—
5	Simple renal cysts	—

ADPKD — illness script

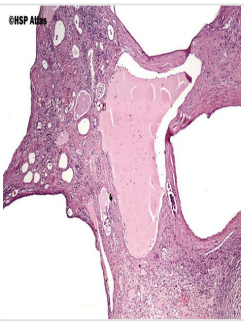
WHO	Common **inherited** disorder of the kidney. Manifests clinically between **15–30 years**, although the genetic defect is present at birth. **Autosomal dominant, bilateral.** Severity worsened by: Black ethnicity (sickle cell trait), male sex, hypertension.
Genetics	**PKD1** (chromosome 16p13.3) encodes **polycystin-1**, expressed in tubular epithelial cells; accounts for ~85% of mutations. <5% develop renal failure by 40 years, rising to >70% at 60–70 years. **PKD2** (chromosome 4q21) encodes **polycystin-2**, expressed in all renal tubule segments and many extrarenal tissues. PKD2 is **less severe** than PKD1 (RF <5% at 50 years, rising to 15% at 60 and 45% at 70) — PKD2 patients tend to have older age at onset and later renal failure.
Clinical	Pain, hypertension, hematuria, palpable renal masses, progression to renal failure. Extrarenal: polycystic liver, pancreas, spleen; intracranial berry aneurysms.
What you see	Gross: kidneys bilaterally enlarged, mass of cysts up to 3–4 cm in diameter. Microscope: cysts filled with clear serous fluid, or turbid red-brown/hemorrhagic fluid; cysts arise from tubules and so have variable lining epithelia.



ADPKD — bilaterally enlarged, cystic kidney



Cysts arising from tubules, variable lining



Cyst wall detail

ARPKD — illness script

WHO	**Rare**; multiple bilateral cysts present **at birth**.
Genetics	**PKHD1** gene (chromosome 6p21-p23) encodes protein **fibrocystin**, also expressed in liver and pancreas. Function unknown — may be a cell-surface receptor with a role in collecting duct and biliary differentiation.
Association	May develop **Congenital Hepatic Fibrosis (CHF)**: periportal fibrosis and proliferation of biliary ductules.
What you see	Gross: kidneys enlarged with **smooth external surface**; on cut section numerous small cysts at right angles to the cortical surface. Microscope: cylindrical or saccular dilations; cysts have a **uniform lining of cuboidal cells**.



ARPKD — enlarged kidney, smooth surface



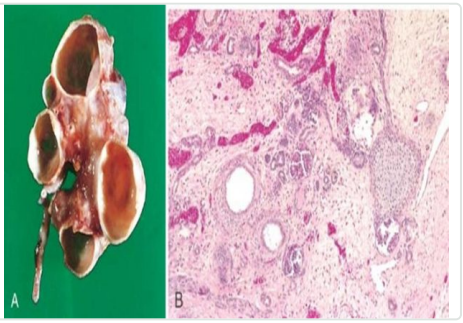
Uniform cuboidal-lined cylindrical/saccular cysts

ADPKD vs ARPKD — the discriminator

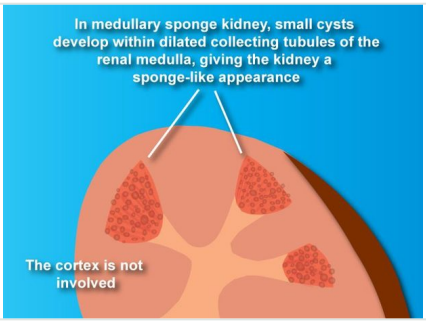
Feature	ADPKD	ARPKD
Inheritance	Autosomal dominant	Autosomal recessive
Onset	Clinically 15–30y (defect present at birth)	Multiple bilateral cysts already at birth; rare
Gene / protein	PKD1/polycystin-1 (16p13.3) or PKD2/ polycystin-2 (4q21)	PKHD1/fibrocystin (6p21-p23)
Extrarenal	Liver/pancreas/spleen cysts, berry aneurysms	Congenital hepatic fibrosis
Gross surface	Enlarged, irregular, cysts up to 3–4 cm	Enlarged, **smooth** external surface, small cysts at right angles to cortex
Cyst lining	Variable (cysts arise from tubules generally)	Uniform cuboidal (cylindrical/saccular)

The rest of the cystic diseases — one-liners

Entity	Key features
Multicystic renal dysplasia	Sporadic; abnormality in metanephric differentiation; uni- or bilateral. Gross: kidney enlarged, irregular, multicystic. Microscope: cysts lined by flattened epithelium — characteristic feature is **islands of undifferentiated mesenchyme, or cartilage, and immature collecting ducts**.
Medullary sponge kidney	Multiple small medullary cysts, **usually without renal failure**; renal function normal. Adults; usually an incidental radiographic finding, or complicated by infection. Pathogenesis unknown. Gross: dilated papillary ducts in medulla. Microscope: cysts lined by cuboidal, occasionally transitional, epithelium.
Nephronophthisis	Cysts in the medulla usually at the **corticomedullary junction**; cortical tubulointerstitial damage causes renal insufficiency. Progressive. Three variants: sporadic non-familial, **familial juvenile** (most common), renal-retinal dysplasia (with ocular lesion). M/F: cysts lined by flattened/cuboidal epithelium; glomeruli generally preserved.
Acquired (dialysis-associated) cystic disease	Associated with long-term dialysis. Multiple cysts, 0.1–4 cm; glomerular and tubular atrophy and scarring. Cysts contain clear fluid, lined by hyperplastic or flattened tubular epithelium. Mostly asymptomatic, but may cause hematuria. **Up to 100-fold increased risk of renal cell carcinoma.**
Simple renal cysts	**Most common renal cyst**; single or multiple cortical cysts in adults, 1–5 cm (may reach 10 cm+). Often asymptomatic; hemorrhage may cause sudden distention and pain. Translucent, lined by a glistening smooth membrane, filled with clear fluid. Microscope: single layer of cuboidal or flattened epithelium. Main importance: differentiating from kidney tumours — smooth contours, avascular, fluid (not solid) signal on ultrasonography.



Multicystic renal dysplasia — gross & microscopic



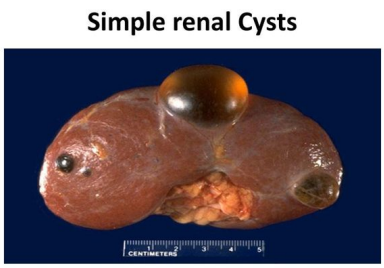
Medullary sponge kidney — cysts confined to medulla, cortex spared



Medullary sponge kidney — gross



Nephronophthisis — small corticomedullary cysts with chronic TIN



Simple renal cyst — translucent, clear fluid

P Must-remember pearls

- Radiopacity splits the two commonest stones: **calcium = radiopaque**, **struvite = radiolucent**.
- Struvite (ammonium magnesium phosphate) stones need **urease-positive bacteria** (classically *Proteus*) and alkaline urine — they are the ones that form **staghorn calculi**.
- **PKD1** mutations are both more common (~85%) and more severe (earlier renal failure) than PKD2.
- ARPKD kidneys have a **smooth** external surface with cysts at right angles to the cortex — the opposite look to the irregular, grossly cystic ADPKD kidney.
- The one histologic feature that identifies multicystic renal dysplasia: **islands of undifferentiated mesenchyme or cartilage**.
- Acquired (dialysis-associated) cystic disease carries up to a **100-fold increased risk of renal cell carcinoma**.
- Simple renal cysts matter mainly because they must be told apart from renal tumours: smooth, avascular, fluid (not solid) on ultrasound.

Q Test yourself

1. A urine culture grows Proteus vulgaris and the urine pH is persistently alkaline. A large staghorn calculus is seen on imaging. Which stone type is this?

A. Calcium oxalate
B. Ammonium magnesium phosphate (struvite)
C. Uric acid
D. Cystine

2. A neonate is found to have bilaterally enlarged kidneys with a smooth external surface; cut section shows numerous small cysts at right angles to the cortex. This is most consistent with:

A. ADPKD
B. ARPKD
C. Multicystic renal dysplasia
D. Simple renal cysts

3. Renal biopsy in a cystic kidney shows islands of undifferentiated mesenchyme and cartilage alongside flattened-epithelium cysts. This finding is characteristic of:

A. Medullary sponge kidney
B. Nephronophthisis
C. Multicystic renal dysplasia
D. Acquired cystic disease

4. A 55-year-old on long-term hemodialysis develops new hematuria. Imaging shows multiple small renal cysts. The main concern specific to this diagnosis is:

A. Progression to ADPKD
B. Up to a 100-fold increased risk of renal cell carcinoma
C. Congenital hepatic fibrosis
D. Struvite stone formation

5. A patient found to have PKD2 mutation asks how their prognosis compares with PKD1. The most accurate answer is that PKD2 disease is:

A. More severe, with earlier renal failure
B. Less severe, with later-onset renal failure
C. Identical in severity and onset
D. Not associated with renal failure at all

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 · B — Urease-positive organisms (Proteus) and alkaline urine are the signature of struvite (ammonium magnesium phosphate) stones, which form staghorn calculi.

2 · B — A smooth external surface with cysts at right angles to the cortex, at birth, is the described gross appearance of ARPKD.

3 · C — Islands of undifferentiated mesenchyme or cartilage are the characteristic histologic feature of multicystic renal dysplasia.

4 · B — Acquired (dialysis-associated) cystic disease carries up to a 100-fold increased risk of renal cell carcinoma.

5 · B — PKD2 is less severe than PKD1, with later-onset renal failure (15% at 60y and 45% at 70y, vs much higher rates for PKD1).

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam

Pathology GUE Block · MEDucation by Mattin Majid · summarised from SGL 1 (Tubular and tubulointerstitial diseases)
Study alongside the lecture, not instead of it.